The mass of the Earth's atmosphere is 5×10^{18} kilograms, of which about 80 percent is nitrogen and 20 percent is molecular oxygen, O_2—that is, 1×10^{18} kilograms of O_2. The amount of carbon in undecayed plant life (called "organic carbon" by geophysicists) is about 3×10^{15} kilograms, with roughly half in the oceans' surface layers and half on land (Table 1 of Hedges and Keil [1995]). Both forms get oxidized (converted to CO_2) in about thirty years on average. Since CO_2 has two oxygen atoms (that come from the atmosphere) and just one carbon atom, and the mass of each oxygen atom is 16/12 that of a carbon atom, the oxidization of all this carbon, after all plants die, would eat up $2 \times 16/12 \times (3 \times 10^{15}$ kilograms$) = 1 \times 10^{16}$ kilograms of O_2, which is 1 percent of the atmosphere's oxygen.

For evidence of sudden overturns of the Earth's oceans and the theory of how they might be produced, see Adkins, Ingersoll, and Pasquero (2005). The standard estimate of the amount of organic carbon in sediments on the ocean bottoms that might be brought to the surface by such an overturn focuses on an upper sedimentary layer that is mixed by ocean currents and animal activity. This mixed layer's carbon content is the product of an estimated rate of deposit of carbon into the sediments (about 1011 kilograms per year) and the average time it takes for its carbon to be oxidized by oxygen from ocean water (1000 years), giving 1.5×10^{14} kilograms, one-twentieth of that on land and in ocean surface layers (Emerson and Hedges 1988, Hedges and Keil 1995). However: (i) The estimated deposition rate could be wrong by a huge amount; for example, Baumgart et al. (2009), relying on extensive measurements, estimate a deposition rate in the Indian Ocean off Java and Sumatra that is uncertain by a factor of fifty and, extrapolated to the whole ocean could give as much as 3×10^{15} kilograms of carbon in the mixed layer (the same as on land and in the ocean's surface layers). (ii) A substantial fraction of the deposited carbon could sink into a lower